

## **PART A**

### **Question 1**

|                              | <b>Predictive Positive (Disease)</b> | <b>Predictive Negative (No Disease)</b> |
|------------------------------|--------------------------------------|-----------------------------------------|
| Actual Positive (Disease)    | 140 (TP)                             | 10 (FN)                                 |
| Actual Negative (No Disease) | 30 (FP)                              | 20 (TN)                                 |

#### **1) Accuracy**

$$\begin{aligned} &= (TP + TN) / \text{Total} \\ &= (140 + 20) / 200 \\ &= \mathbf{0.80 \text{ or } 80\%} \end{aligned}$$

#### **2) Misclassification Rate**

$$\begin{aligned} &= (FP + FN) / \text{Total} \\ &= (30 + 10) / 200 \\ &= \mathbf{0.20 \text{ or } 20\%} \end{aligned}$$

#### **3) Positive Precision Value (PPV)**

$$\begin{aligned} &= TP / (TP + FP) \\ &= 140 / (140 + 30) \\ &= \mathbf{0.8235 \text{ or } 82.35\%} \end{aligned}$$

#### **4) True Positive Call (TPR)**

$$\begin{aligned} &= TP / (TP + FN) \\ &= 140 / (140 + 10) \\ &= \mathbf{0.9333 \text{ or } 93.33\%} \end{aligned}$$

#### **5) Interpretation:**

The model has high sensitivity, meaning it correctly identifies most patients with the disease. However, precision is slightly lower, indicating some false positives.

## Question 2

|                 | Predictive Spam | Predictive Non-Spam |
|-----------------|-----------------|---------------------|
| Actual Spam     | 180 (TP)        | 20 (FN)             |
| Actual Non-Spam | 30 (FP)         | 70 (TN)             |

### 1) Accuracy

$$\begin{aligned} &= (TP + TN) / \text{Total} \\ &= (180 + 70) / 300 \\ &= \mathbf{0.8333 \text{ or } 83.33\%} \end{aligned}$$

### 2) Misclassification Rate

$$\begin{aligned} &= (FP + FN) / \text{Total} \\ &= (30 + 20) / 300 \\ &= \mathbf{0.1667 \text{ or } 16.67\%} \end{aligned}$$

### 3) Positive Precision Value (PPV)

$$\begin{aligned} &= TP / (TP + FP) \\ &= 180 / (180 + 30) \\ &= \mathbf{0.8571 \text{ or } 85.71\%} \end{aligned}$$

### 4) True Positive Call (TPR)

$$\begin{aligned} &= TP / (TP + FN) \\ &= 180 / (180 + 20) \\ &= \mathbf{0.90 \text{ or } 90\%} \end{aligned}$$

### 5) Interpretation:

The model performs well with true positive call, which means it accurately detects spam emails and minimizes both false positives and false negatives.

### Question 3

|              | Predictive Churn | Predictive Loyal |
|--------------|------------------|------------------|
| Actual Churn | 80 (TP)          | 10 (FN)          |
| Actual Loyal | 20 (FP)          | 90 (TN)          |

**1) Accuracy**

$$= (TP + TN) / \text{Total}$$

$$= (80 + 90) / 200$$

$$= \mathbf{0.85 \text{ or } 85\%}$$

**2) Misclassification Rate**

$$= (FP + FN) / \text{Total}$$

$$= (20 + 10) / 200$$

$$= \mathbf{0.15 \text{ or } 15\%}$$

**3) Positive Precision Value (PPV)**

$$= TP / (TP + FP)$$

$$= 80 / (80 + 20)$$

$$= \mathbf{0.80 \text{ or } 80\%}$$

**4) True Positive Call (TPR)**

$$= TP / (TP + FN)$$

$$= 80 / (80 + 10)$$

$$= \mathbf{0.8889 \text{ or } 88.89\%}$$

**5) Interpretation:**

This model effectively identifies churned customers with high accuracy, precision, and recall. It's reliable for customer retention strategies.

## **Part B**

| <b>Num.</b> | <b>Item</b>             | <b>Description</b>                                                                              |
|-------------|-------------------------|-------------------------------------------------------------------------------------------------|
| 1           | Overfitting             | When a model learns noise/details from training data too well, performing poorly on new data.   |
| 2           | Underfitting            | When a model is too simple and fails to capture the underlying pattern of the data.             |
| 3           | Hold-out Validation     | Splitting data into training and testing sets (e.g., 80% train, 20% test) for model evaluation. |
| 4           | K-fold Cross Validation | Data is split into $k$ parts; each part is used as test set once while training on the others.  |
| 5           | Training set            | Portion of the dataset used to train the machine learning model.                                |
| 6           | Testing set             | Portion of the dataset used to evaluate the trained model's performance.                        |
| 7           | Target variable/Label   | The output or result the model aims to predict (e.g., disease, spam, churn).                    |
| 8           | Classification          | Predicting categorical labels (e.g., yes/no, spam/not spam, disease/no disease).                |
| 9           | Regression              | Predicting continuous values (e.g., price, temperature, age).                                   |
| 10          | Stratified sampling     | Ensuring each class is proportionally represented in training/testing splits.                   |